Exercise #6

20. February 2023

Problem 1.

Find the Fourier transform of the following functions $f\colon \mathbb{R} \to \mathbb{R}$:

a) The function

$$f(x) = \mathrm{e}^{-2|x|}.$$

b) The function

$$f(x) = \begin{cases} e^x, & x \in [-a, a], \\ 0, & \text{otherwise}, \end{cases}$$

for some $a > 0$.

c) The function

$$f(x) = x^2 \mathrm{e}^{-x^2}.$$

Hint: Differentiate e^{-x^2} twice, and use the table in Kreyszig, p. 536.

Solution.

a) We compute directly,

$$\begin{aligned}\sqrt{2\pi}\mathcal{F}(e^{-2|x|}) &= \int_{\mathbb{R}} e^{-2|x|-i\omega x}dx = \int_{-\infty}^{0} e^{2x-i\omega x}dx + \int_{0}^{\infty} e^{-2x-i\omega x}dx \\ &= \left[\frac{e^{2x-i\omega x}}{2-i\omega}\right]_{-\infty}^{0} + \left[\frac{e^{-2x-i\omega x}}{-2-i\omega}\right]_{0}^{\infty} = \frac{1}{2-i\omega} + \frac{1}{-2-i\omega} \\ &= \frac{-2i\omega}{-4-\omega^2}.\end{aligned}$$

Therefore,

$$\mathcal{F}(e^{-2|x|}) = \sqrt{\frac{2}{\pi}}\frac{i\omega}{4+\omega^2}$$

b) We compute directly,

$$\sqrt{2\pi}\mathcal{F}(f) = \int_{-a}^{a} e^{x-i\omega x}dx = \left[\frac{e^{x-i\omega x}}{1-i\omega}\right]_{-a}^{a} = \frac{e^{a-i\omega a} + e^{-a+i\omega a}}{1-i\omega}.$$

Therefore,

$$\mathcal{F}(f) = \frac{1}{\sqrt{2}}\frac{e^{a-i\omega a} + e^{-a+i\omega a}}{1-i\omega}.$$

c) We use the hint and differentiate e^{-x^2} twice. This gives,

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\mathrm{e}^{-x^2} = 4x^2\mathrm{e}^{-x^2} - 2\mathrm{e}^{-x^2}.$$

Hence,

$$x^2\mathrm{e}^{-x^2} = \frac{\mathrm{d}^2}{\mathrm{d}x^2}\frac{1}{4}\mathrm{e}^{-x^2} + \frac{1}{2}\mathrm{e}^{-x^2}.$$

Fourier transforming, we get,

$$\begin{aligned}\mathcal{F}(x^2\mathrm{e}^{-x^2}) &= \mathcal{F}\left(\frac{d^2}{dx^2}\frac{1}{4}\mathrm{e}^{-x^2}\right) + \mathcal{F}\left(\frac{1}{2}\mathrm{e}^{-x^2}\right) \\ &= \frac{1}{4}(-\omega^2)\mathcal{F}(\mathrm{e}^{-x^2}) + \frac{1}{2}\mathcal{F}(\mathrm{e}^{-x^2}) \\ &= \left(\frac{1}{2} - \frac{\omega^2}{4}\right)\frac{\mathrm{e}^{-\frac{\omega^2}{4}}}{\sqrt{2}}.\end{aligned}$$

Here we used that

$$\mathcal{F}(\mathrm{e}^{-x^2}) = \frac{\mathrm{e}^{-\frac{\omega^2}{4}}}{\sqrt{2}},$$

which can be shown using a similar approach as in problem 1.

Problem 2.

Define the functions $f, g\colon \mathbb{R} \to \mathbb{R}$ as

$$f(x) := e^{-x^2} \quad \text{and} \quad g(x) := e^{-2x^2}.$$

Use the Fourier transform in order to compute the convolution $f * g$.

You may (and should) use the table in Kreyszig, p. 536, for the computation of the Fourier transform of e^{-ax^2}.

Solution.

We use the formula from Kreyszig

$$\mathcal{F}\left(e^{-ax^2}\right) = \frac{1}{\sqrt{2a}} e^{-\frac{\omega^2}{4a}}$$

with $a = 1$ and $a = 2$ and obtain that

$$\hat{f}(\omega) = \frac{1}{\sqrt{2}} e^{-\frac{\omega^2}{4}} \qquad \text{and} \qquad \hat{g}(\omega) = \frac{1}{2} e^{-\frac{\omega^2}{8}}.$$

Now the convolution theorem states that

$$\mathcal{F}(f * g)(\omega) = \sqrt{2\pi} \hat{f}(\omega) \cdot \hat{g}(\omega) = \sqrt{2\pi} \frac{1}{\sqrt{2}} \frac{1}{2} e^{-\frac{\omega^2}{4}} e^{-\frac{\omega^2}{8}} = \frac{\sqrt{\pi}}{2} e^{-\frac{3\omega^2}{8}}.$$

In order to obtain the convolution $(f * g)(x)$, we thus have to find the inverse Fourier transform of the function

$$\hat{h}(\omega) = \frac{\sqrt{\pi}}{2} e^{-\frac{3\omega^2}{8}}.$$

To that end, we again apply the formula from Kreyszig with a suitable parameter a. By looking at the exponent in the function $\hat{h}$, we see that the correct choice is $a = \frac{2}{3}$. Now the formula implies that

$$\mathcal{F}^{-1}\Big(\frac{1}{\sqrt{4/3}}e^{-\frac{3\omega^2}{8}}\Big)(x) = e^{-\frac{2x^2}{3}}.$$

Multiplying this equation by $\frac{\sqrt{\pi}}{2}\frac{2}{\sqrt{3}} = \frac{\sqrt{\pi}}{\sqrt{3}}$ and using the linearity of the Fourier transform, we obtain that

$$h(x) = \mathcal{F}^{-1}\Big(\frac{\sqrt{\pi}}{2}e^{-\frac{3\omega^2}{8}}\Big)(x) = \frac{\sqrt{\pi}}{\sqrt{3}}\mathcal{F}^{-1}\Big(\frac{\sqrt{3}}{2}e^{-\frac{3\omega^2}{8}}\Big)(x) = \frac{\sqrt{\pi}}{\sqrt{3}}e^{-\frac{2x^2}{3}}.$$

Thus the convolution of f and g is

$$(f * g)(x) = h(x) = \frac{\sqrt{\pi}}{\sqrt{3}}e^{-\frac{2x^2}{3}}.$$

Problem 3.

Define the functions $f, g \colon \mathbb{R} \to \mathbb{R}$ as

$$f(x) := \begin{cases} 1 & \text{if } 0 < x < 1, \\ 0 & \text{else,} \end{cases} \quad \text{and} \quad g(x) := \begin{cases} x & \text{if } 0 < x \leq 1, \\ 2 - x & \text{if } 1 < x < 2, \\ 0 & \text{else.} \end{cases}$$

a) Use the definition of convolution in order to show that $f * f = g$.

b) Compute the Fourier transform of f.

c) Compute the Fourier transform of g.

Solution.

a) The definition of convolution states that

$$(f * f)(x) = \int_{-\infty}^{+\infty} f(y)f(x-y)\,dy.$$

For our particular case, the function $f(x-y)$ is equal to 1 for $0 < x-y < 1$, that is, for values $x-1 < y < x$. Thus

$$(f * f)(x) = \int_{x-1}^{x} f(y)\,dy.$$

We thus have four different cases:

- If $x \leq 0$, then the upper limit of integration is smaller than 0, which means that the integrand $f(y)$ is always equal to 0. Thus $(f * f)(x) = 0$ in this case.

- If $0 < x \leq 1$, then the lower limit of integration is smaller than 0, but the upper limit is between 0 and 1. The integral thus reduces to

$$(f * f)(x) = \int_{x-1}^{x} f(y)\,dy = \int_{0}^{x} 1\,dy = x.$$

- If $1 < x \leq 2$, then the lower limit of integration is between 0 and 1, but the upper limit is larger than 1. Then the integral becomes

$$(f * f)(x) = \int_{x-1}^{x} f(y)\,dy = \int_{x-1}^{1} 1\,dy = 2 - x.$$

- Finally, if $x > 2$, then the lower limit of integration is larger than 1, and we obtain again that $(f * f)(x) = 0$.

Putting everything together, we see that

$$(f * f)(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ x & \text{if } 0 < x \leq 1, \\ 2 - x & \text{if } 1 < x \leq 2, \\ 0 & \text{if } x > 2, \end{cases}$$

which is exactly the function g.

b) For the Fourier transform of f, we can simply use the definition:

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x) e^{-i\omega x}\,dx = \frac{1}{\sqrt{2\pi}} \int_{0}^{1} e^{-i\omega x}\,dx.$$

For the case $\omega = 0$, we obtain

$$\hat{f}(0) = \frac{1}{\sqrt{2\pi}} \int_0^1 1\,dx = \frac{1}{\sqrt{2\pi}}.$$

For the case $\omega \neq 0$, we obtain

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^1 e^{-i\omega x}\,dx = -\frac{1}{\sqrt{2\pi}} \frac{e^{-i\omega x}}{i\omega}\bigg|_0^1 = -\frac{1}{\sqrt{2\pi}} \frac{e^{-i\omega} - 1}{i\omega} = \frac{1 - e^{-i\omega}}{\sqrt{2\pi}i\omega}.$$

If we want (we don't have to), we can rewrite this as

$$\frac{1 - e^{-i\omega} - 1}{\sqrt{2\pi}i\omega} = e^{i\frac{\omega}{2}} \frac{e^{i\frac{\omega}{2}} - e^{-i\frac{\omega}{2}}}{\sqrt{2\pi}i\omega} = e^{i\frac{\omega}{2}} \frac{\sqrt{2}}{\sqrt{\pi}} \operatorname{sinc}(\omega/2).$$

c) Since we have already computed $\hat{f}$ and we note that $g = f * f$, the simplest way is to use the convolution theorem. Thus

$$\hat{g}(\omega) = \sqrt{2\pi}(\hat{f}(\omega))^2 = -\frac{(1 - e^{-i\omega})^2}{\sqrt{2\pi}\omega^2}.$$

Problem 4.

a) Let $w_N := \mathrm{e}^{-\frac{2\pi \mathrm{i}}{N}}$ for some $N \in \mathbb{N}$. Show that,

$$\sum_{k=0}^{N-1} w_N^{jk} = \begin{cases} N, & \text{if } j \bmod N = 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $j \in \mathbb{Z}$ is some integer.

b) Show that,

$$\sum_{k=0}^{N-1} e^{\frac{2\pi i n}{N}k} e^{-\frac{2\pi i m}{N}k} = \begin{cases} N, & n = m, \\ 0, & \text{otherwise,} \end{cases}$$

for some $n, m \in \{0, \dots, N-1\}$.

c) Compute explicitly the Fourier matrices $\mathcal{F}_2$, $\mathcal{F}_3$, and $\mathcal{F}_4$.

d) Compute by hand the discrete Fourier transform of the vector $f = (1, 2, 3, 3)$.

e) Assume that the discrete Fourier transform of the vector $f \in \mathbb{R}^{11}$ is given as

$$\hat{f} = (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1).$$

Express the entries f_n of f in terms of sine and cosine functions.

Solution.

a) First we note that if $j \bmod N = 0$, then we have $j = m \cdot N$, for some $m \in \mathbb{Z}$. Therefore,

$$w_N^j = e^{-\frac{2\pi ji}{N}} = e^{-2\pi mi} = (e^{-2\pi \mathrm{i}})^m = 1^m = 1.$$

It follows that $\sum_{k=0}^{N-1} w_N^{jk} = \sum_{k=0}^{N-1} 1^k = N$.

When $j \bmod N \neq 0$, we have, using the formula for geometric sums,

$$\sum_{k=0}^{N-1} w_N^{jk} = \frac{1 - w_N^{Nj}}{1 - w_N^j} = 0$$

because $w_N^{Nj} = e^{-\frac{2\pi jiN}{N}} = e^{-2\pi ji} = 1$.

b) Here we note that,

$$\sum_{k=0}^{N-1} e^{\frac{2\pi n}{N}k} e^{-\frac{2\pi im}{N}k} = \sum_{k=0}^{N-1} e^{\frac{2\pi \mathrm{i}(n-m)}{N}k}.$$

When $n = m$, this is clearly equal to N, as all terms in the sum are equal to 1.

To show that it is zero when $n \neq m$, we can use the formula for a geometric sums. Alternatively, we can use the complicated expression from the first point: since $n - m \bmod N \neq 0$ when $n \neq m$ and $n, m \in \{0, 1, \ldots, N-1\}$, we have from the first problem that the sum is 0 in this case.

c) The entries of the Fourier matrix $\mathcal{F}_N$ are w_N^{kn}, where $w_N = e^{-i\frac{2\pi}{N}}$.

- For $N = 2$, we have $w_2 = -1$ and thus

$$\mathcal{F}_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

- For $N = 3$, we have

$$w_3 = e^{-i\frac{2\pi}{3}} = \cos\left(\frac{2\pi}{3}\right) - i\sin\left(\frac{2\pi}{3}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

Thus

$$\mathcal{F}_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & \frac{1}{2} - \frac{\sqrt{3}}{2}i & \frac{1}{2} + \frac{\sqrt{3}}{2}i \\ 1 & \frac{1}{2} + \frac{\sqrt{3}}{2}i & \frac{1}{2} - \frac{\sqrt{3}}{2}i. \end{pmatrix}$$

- For $N = 4$, we have $w_4 = -i$, and thus

$$\mathcal{F}_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}.$$

d) We have

$$\hat{f} = \mathcal{F}_4 f = \mathcal{F}_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ i-2 \\ -1 \\ -2-i \end{pmatrix}.$$

e) Here we have to recall that the rows of the matrix $\mathcal{F}_{11}$ are all orthogonal in inner product on $\mathbb{C}^{11}$. So if

$$\mathcal{F}_{11} f = (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1),$$

we must actually have, $f = \frac{1}{11}(w_1 + w_{10})$, where,

$$w_k = \{e^{-\frac{2\pi i k}{11}n}\}_{n=0}^{10}.$$

(Check for yourself that $\langle w_i, w_j \rangle = 11\delta_{i,j}$).

To express this in terms of sine and cosine functions, we must recall that,

$$e^{xi} = \cos(x) + i\sin(x),$$

which yields,

$$\begin{aligned} 11f_n &= e^{-\frac{2\pi i}{11}n} + e^{-\frac{2\pi i}{11}10n} \\ &= e^{-\frac{2\pi i}{11}n} + e^{-2\pi i n + \frac{2\pi i}{11}n} \\ &= e^{-\frac{2\pi i}{11}n} + e^{-2\pi i n}e^{\frac{2\pi i}{11}n} \\ &= e^{-\frac{2\pi i}{11}n} + e^{\frac{2\pi i}{11}n} \\ &= 2\cos(\frac{2}{11}\pi n). \end{aligned}$$

The next exercises are optional and should not be handed in!
Problem 5.

Compute the convolution $f * f$, where $f\colon \mathbb{R} \to \mathbb{R}$ is the function

$$f(x) = \operatorname{sinc}(x) := \begin{cases} \dfrac{\sin(x)}{x} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$$

Hint: Use the Fourier transform!

Solution.

The trick here is to realise that f is the inverse Fourier transform of a rectangle pulse: Let

$$\hat{g}(\omega) = \begin{cases} 1 & \text{if } -1 < \omega < 1, \\ 0 & \text{else.} \end{cases}$$

Then (ignoring the problems at $x = 0$),

$$\begin{aligned} g(x) = \mathcal{F}^{-1}(\hat{g})(x) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{i\omega x}\, d\omega = \frac{1}{\sqrt{2\pi}} \frac{e^{i\omega x}}{ix}\bigg|_{\omega=-1}^{1} \\ = \frac{e^{ix} - e^{-ix}}{\sqrt{2\pi} ix} = \sqrt{\frac{2}{\pi}} \frac{\sin(x)}{x} = \sqrt{\frac{2}{\pi}} \operatorname{sinc}(x). \end{aligned}$$

This implies that

$$\hat{f}(\omega) = \sqrt{\frac{\pi}{2}} \hat{g}(\omega).$$

Now the convolution theorem implies that

$$\mathcal{F}(f * f)(\omega) = \sqrt{2\pi} \hat{f}(\omega) \cdot \hat{f}(\omega) = \frac{\pi^{3/2}}{\sqrt{2}} \hat{g}(\omega)^2.$$

However, by the definition of $\hat{g}(\omega)$, we have

$$\hat{g}(\omega)^2 = \hat{g}(\omega).$$

Thus

$$\mathcal{F}(f * f)(\omega) = \frac{\pi^{3/2}}{\sqrt{2}} \hat{g}(\omega).$$

Applying the inverse Fourier transform on both sides implies that

$$(f * f)(x) = \frac{\pi^{3/2}}{\sqrt{2}} g(x) = \pi \operatorname{sinc}(x).$$

Problem 6.

Compute the Fourier transform of the following functions $f\colon \mathbb{R} \longrightarrow \mathbb{R}$:

a) The function

$$f(x) = xe^{-|x|}.$$

b) The function

$$f(x) = \begin{cases} \sin(x) & \text{if } 0 < x < \pi, \\ 0 & \text{else.} \end{cases}$$

Solution.

a) From the definition of the Fourier transform we get

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} xe^{-|x|} e^{-i\omega x}\, dx.$$

We split up this integral to one over the positive real axis, and one over the negative real axis. For the former, we get

$$\begin{aligned} \int_0^\infty xe^{-x} e^{-i\omega x}\, dx = \int_0^\infty xe^{-(1+i\omega)x}\, dx &= \left.\frac{xe^{-(1+i\omega)x}}{-(1+i\omega)}\right|_0^\infty + \int_0^\infty \frac{e^{-(1+i\omega)x}}{1+i\omega}\, dx \\ &= 0 - \left.\frac{e^{-(1+i\omega)x}}{(1+i\omega)^2}\right|_0^\infty = \frac{1}{(1+i\omega)^2}. \end{aligned}$$

Here we have used that

$$\lim_{x\to\infty} xe^{-(1+i\omega)x} = 0 \qquad \text{and} \qquad \lim_{x\to\infty} e^{-(1+i\omega)x} = 0.$$

Similarly, the other integral is

$$\int_{-\infty}^{0} xe^{x}e^{-i\omega x}\,dx = \int_{-\infty}^{0} xe^{(1-i\omega)x}\,dx = \frac{xe^{(1-i\omega)x}}{1-i\omega}\Big|_{0}^{\infty} - \int_{-\infty}^{0}\frac{e^{(1-i\omega)x}}{1-i\omega}\,dx$$
$$= 0 - \frac{e^{(1-i\omega)x}}{(1-i\omega)^2}\Big|_{0}^{\infty} = -\frac{1}{(1-i\omega)^2}.$$

Thus

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}}\left(\frac{1}{(1+i\omega)^2} - \frac{1}{(1-i\omega^2)}\right) = \frac{1}{\sqrt{2\pi}}\frac{(1-i\omega)^2-(1+i\omega)^2}{\big((1+i\omega)(1-i\omega)\big)^2} = -\frac{1}{\sqrt{2\pi}}\frac{4i\omega}{(1+\omega^2)^2}.$$

b) Here we write

$$\sin(x) = \frac{1}{2i}\big(e^{ix}-e^{-ix}\big).$$

Thus

$$\hat{f}(x) = \frac{1}{\sqrt{2\pi}}\int_0^{\pi}\sin(x)e^{-i\omega x}\,dx = \frac{1}{\sqrt{2\pi}}\frac{1}{2i}\int_0^{\pi}\big(e^{ix}-e^{-ix}\big)e^{-i\omega x}\,dx$$
$$= -\frac{i}{2\sqrt{2\pi}}\int_0^{\pi}\left(e^{i(1-\omega)x}-e^{-i(1+\omega)x}\right)dx = -\frac{1}{2\sqrt{2\pi}}\left(\frac{e^{i(1-\omega)x}}{1-\omega}-\frac{e^{-i(1+\omega)x}}{1+\omega}\right)\Bigg|_0^{\pi}$$
$$= -\frac{1}{2\sqrt{2\pi}}\left(\frac{e^{i(1-\omega)\pi}}{1-\omega}-\frac{e^{-i(1+\omega)\pi}}{1+\omega}\right).$$

Problem 7. (See Problem 7 in the exam from spring 2022)

a) Compute by hand the discrete Fourier transform of the vector $f = (1/2, 1, 1/2, 0)$.

b) Let $c \in \mathbb{R}$ be given, and assume that, for some signal $g \in \mathbb{R}^4$ we obtain $\hat{g} = (0, c, 0, c)$. What is the simplest function $g(x)$ that could have been sampled?

c) Is the inverse Fourier transform $h = \mathcal{F}_8^{-1}\hat{h}$ of the vector $\hat{h} = (0,0,0,0,0,0,1,0)$ real-valued?

Solution.

See the solutions to the exam in spring 2022.